

## **Packet Tracer – Ping and Tracert Basics**

### **Objectives**

**Part 1:** Build a Simple Network Topology

**Part 2:** Configure Static IP Addressing

**Part 3:** Verify Connectivity Using Ping

**Part 4:** Explore Network Paths Using Tracert

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### **Background / Scenario**

Network administrators frequently use diagnostic tools to verify whether devices can communicate and to identify where connectivity problems occur.

Two of the most common troubleshooting commands are:

- **ping** — Tests whether a destination device is reachable.
- **tracert** — Displays the path that packets take to reach a destination.

In this lab, you will connect two PCs through a switch, assign static IP addresses, and use both commands to verify connectivity and observe packet routing.

This lab focuses on:

- Configuring static IPv4 addresses
- Testing connectivity with ICMP
- Using ping to verify communication
- Using tracert to view the network path

You are developing essential troubleshooting skills that every network technician and CCNA student must master.

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### **Instructions**

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#### **Part 1: Build the Network Topology**

##### **Step 1: Add the Devices**

- a. Open **Cisco Packet Tracer**.
  - b. From **End Devices**, drag two **PCs** onto the workspace.
  - c. From **Network Devices** → **Switches**, drag one **2960 Switch** onto the workspace.
  - d. Rename the devices:
    - PC0 → **PC-A**
    - PC1 → **PC-B**
    - Switch0 → **S1**
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## **Step 2: Connect the Devices**

Since PCs are connecting to a switch, you will use **Copper Straight-Through** cables.

- a. Click **Connections** (lightning bolt icon).
- b. Select **Copper Straight-Through**.
- c. Connect:
  - PC-A → FastEthernet0
  - S1 → FastEthernet0/1
- d. Connect:
  - PC-B → FastEthernet0
  - S1 → FastEthernet0/2

Wait until all link lights turn **green**.

Your physical topology is now complete.

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## **Part 2: Configure Static IP Addressing**

Because there is no router or DHCP server, you must manually assign IP addresses.

### **Step 1: Configure PC-A**

- a. Click **PC-A**.
- b. Go to the **Desktop** tab.

c. Click **IP Configuration**.

d. Select **Static**.

Enter:

- IP Address: **192.168.1.10**
- Subnet Mask: **255.255.255.0**
- Default Gateway: *(leave blank)*

Close the window.

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## Step 2: Configure PC-B

a. Click **PC-B**.

b. Go to the **Desktop** tab.

c. Click **IP Configuration**.

d. Select **Static**.

Enter:

- IP Address: **192.168.1.20**
- Subnet Mask: **255.255.255.0**
- Default Gateway: *(leave blank)*

Close the window.

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## Why Leave the Default Gateway Blank?

A default gateway is only needed when devices communicate outside their local network.

Since both PCs belong to the same subnet (**192.168.1.0/24**) and communicate through a switch, a router is not required.

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## Part 3: Test Connectivity Using Ping

### Step 1: Ping from PC-A to PC-B

- a. Click **PC-A**.
- b. Go to the **Desktop** tab.
- c. Open **Command Prompt**.
- d. Type:

ping 192.168.1.20

Press **Enter**.

Expected Result:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

You should receive **4 successful replies**.

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## **Step 2: Ping from PC-B to PC-A**

- a. Click **PC-B**.
- b. Open **Command Prompt**.
- c. Type:

ping 192.168.1.10

Press **Enter**.

You should again receive **4 successful replies**.

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## **What Does Ping Do?**

The **ping** command uses the **Internet Control Message Protocol (ICMP)** to send Echo Requests to another device.

If the destination responds with an Echo Reply, the connection is working.

Common ping results include:

| <b>Result</b> | <b>Meaning</b>                   |
|---------------|----------------------------------|
| Reply from... | Destination reached successfully |

| Result                       | Meaning                                 |
|------------------------------|-----------------------------------------|
| Request timed out            | No response received                    |
| Destination host unreachable | The network cannot find the destination |

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## Part 4: Explore the Network Path Using Tracert

### Step 1: Run Tracert from PC-A

a. On **PC-A**, open **Command Prompt**.

b. Type:

```
tracert 192.168.1.20
```

Press **Enter**.

Expected Result:

Tracing route to 192.168.1.20

```
1  <1 ms  <1 ms  <1 ms  192.168.1.20
```

Trace complete.

Because both PCs are on the same local network, only one hop is displayed.

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### Step 2: Observe the Output

The **tracert** command shows each router (hop) that a packet passes through before reaching its destination.

In this lab:

- Hop 1 = PC-B

Since there are no routers between the devices, the path is very short.

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### What Is a Hop?

A **hop** is one stop that a packet makes while traveling across a network.

Example:

PC-A → Switch → Router1 → Router2 → Server

The switch does not count as a hop because it operates at Layer 2.

The routers would appear in the tracert output.

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### Troubleshooting Tips

If the ping or tracert fails:

- ✓ Verify both PCs are connected using **Copper Straight-Through** cables.
  - ✓ Ensure the link lights are green.
  - ✓ Check that both PCs are in the same subnet.
  - ✓ Confirm the IP addresses are entered correctly.
  - ✓ Wait a few seconds after connecting the cables for the interfaces to initialize.
  - ✓ Remember that a switch does not assign IP addresses automatically.
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### What You Learned

- How to configure static IPv4 addresses.
  - How the **ping** command tests network connectivity.
  - How ICMP Echo Requests and Replies work.
  - How the **tracert** command displays the path to a destination.
  - What a network hop represents.
  - Basic troubleshooting techniques for connectivity problems.
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### Challenge Extensions (Optional)

Try modifying the lab:

1. Change PC-B's IP address to:

192.168.2.20

- Does ping still work?
  - What message do you receive?
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2. Add a router between the two PCs and configure different subnets.

Run:

ping <destination IP>

tracert <destination IP>

- How many hops appear now?
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3. Disconnect one cable and run ping again.

- What error message is displayed?
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4. Add a third PC connected to the switch and test connectivity between all devices.